

House Price Prediction Using Linear Regression

1. Introduction

This project aims to predict house prices using the California Housing Dataset and a Linear Regression model. The objective is to understand the complete machine learning workflow, including data loading, preprocessing, exploratory data analysis (EDA), model training, and evaluation.

2. Dataset Description

The California Housing Dataset was used for this project.

Dataset Information

- Total Records: 20,640
- Features: 8
- Target Variable: MedHouseVal (Median House Value)

Features Used

- MedInc
- HouseAge
- AveRooms
- AveBedrms
- Population
- AveOccup
- Latitude
- Longitude

The dataset contained no missing values.

3. Exploratory Data Analysis (EDA)

EDA was performed to understand the dataset.

Findings

- No missing values were found.
- Median Income (MedInc) showed the strongest positive correlation with house prices.

- Geographic features such as Latitude and Longitude also influenced house values.
- Feature distributions and correlation heatmaps were generated for analysis.

```
df.info()
```

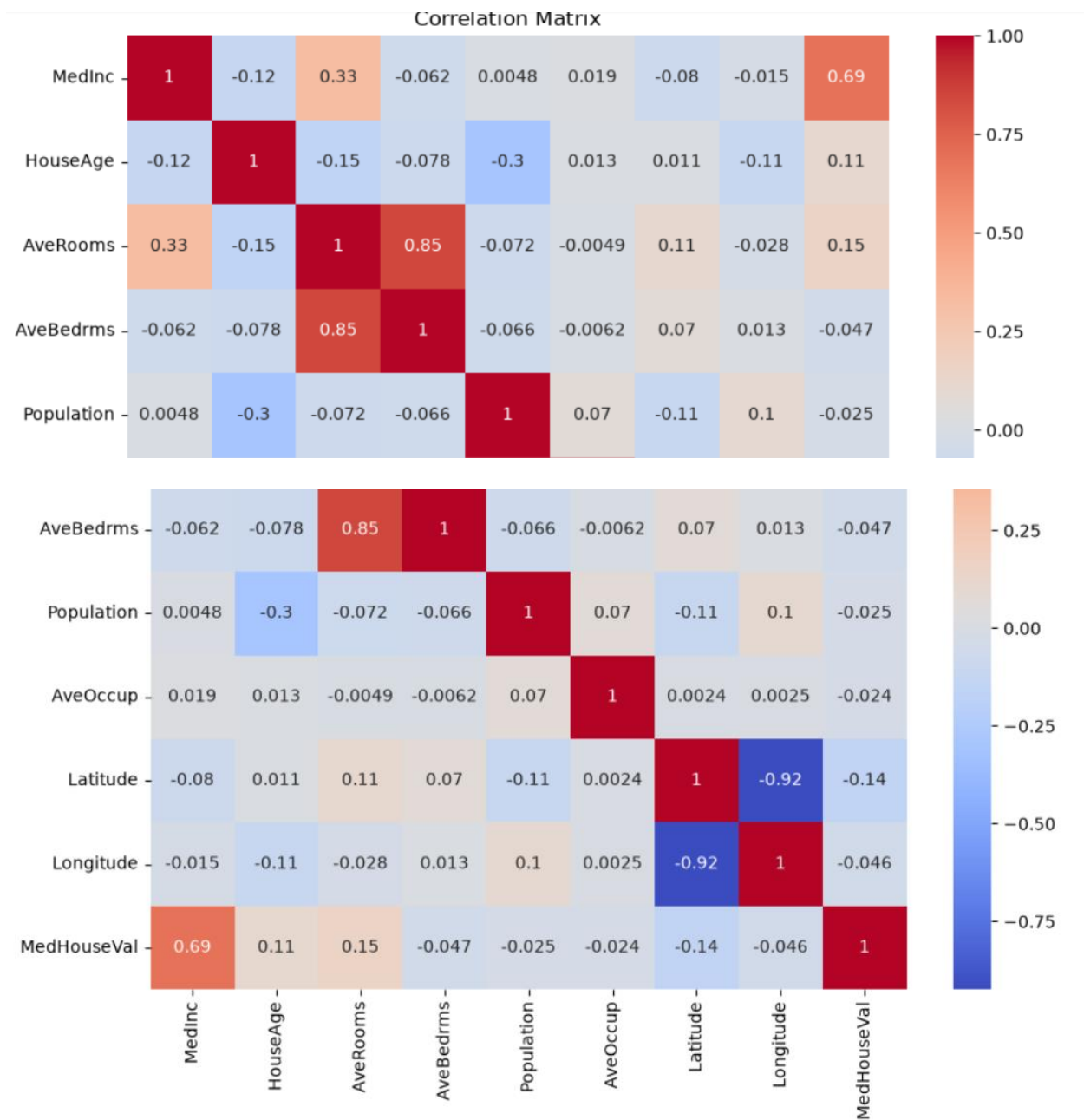
```
<class 'pandas.DataFrame'>
RangeIndex: 20640 entries, 0 to 20639
Data columns (total 9 columns):
#   Column          Non-Null Count  Dtype
---  -
0   MedInc          20640 non-null  float64
1   HouseAge        20640 non-null  float64
2   AveRooms        20640 non-null  float64
3   AveBedrms       20640 non-null  float64
4   Population      20640 non-null  float64
5   AveOccup        20640 non-null  float64
6   Latitude        20640 non-null  float64
7   Longitude       20640 non-null  float64
8   MedHouseVal     20640 non-null  float64
dtypes: float64(9)
memory usage: 1.4 MB
```

```
df.describe()
```

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	MedHouseVal
count	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000
mean	3.870671	28.639486	5.429000	1.096675	1425.476744	3.070655	35.631861	-119.569704	2.068558
std	1.899822	12.585558	2.474173	0.473911	1132.462122	10.386050	2.135952	2.003532	1.153956
min	0.499900	1.000000	0.846154	0.333333	3.000000	0.692308	32.540000	-124.350000	0.149990
25%	2.563400	18.000000	4.440716	1.006079	787.000000	2.429741	33.930000	-121.800000	1.196000
50%	3.534800	29.000000	5.229129	1.048780	1166.000000	2.818116	34.260000	-118.490000	1.797000
75%	4.743250	37.000000	6.052381	1.099526	1725.000000	3.282261	37.710000	-118.010000	2.647250
max	15.000100	52.000000	141.909091	34.066667	35682.000000	1243.333333	41.950000	-114.310000	5.000010

```
df.isnull().sum()
```

```
MedInc      0
HouseAge    0
AveRooms    0
AveBedrms   0
Population  0
AveOccup    0
Latitude    0
Longitude   0
MedHouseVal 0
dtype: int64
```



4. Model Development

Machine Learning Algorithm

Linear Regression

Data Splitting

- Training Data: 80%
- Testing Data: 20%

The dataset was split using `train_test_split()` with `random_state=42`.

The Linear Regression model was trained using the training dataset and predictions were generated on the test dataset.

5. Model Evaluation

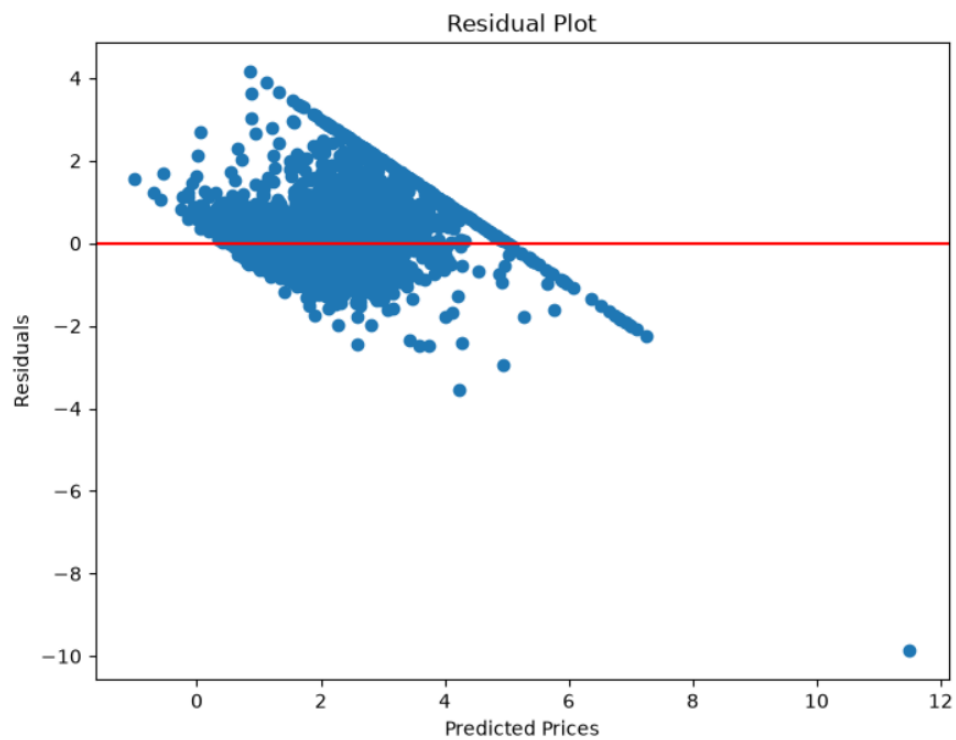
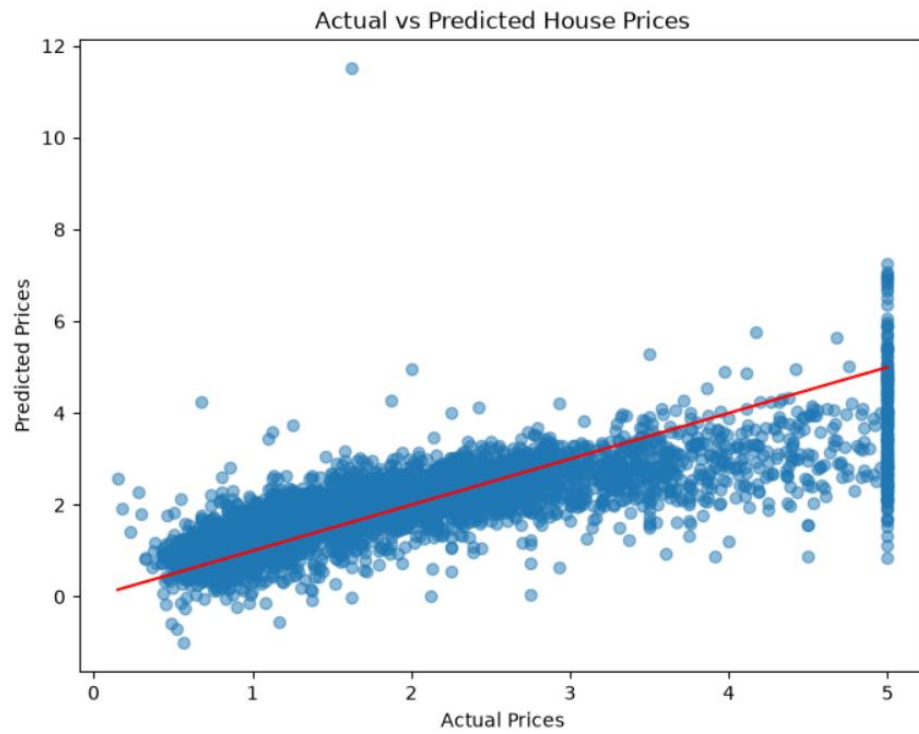
The model was evaluated using MAE, RMSE, and R^2 Score.

Results

Metric	Value
MAE	0.533
RMSE	0.746
R^2 Score	0.576

Interpretation

- MAE of 0.533 indicates a moderate prediction error.
- RMSE of 0.746 indicates the average prediction deviation.
- R^2 Score of 0.576 means the model explains approximately 57.6% of the variance in house prices.



6. Conclusion

A Linear Regression model was successfully developed to predict California house prices. The model achieved an R^2 score of approximately 57.6%, indicating moderate predictive performance.

This project demonstrated the complete machine learning lifecycle including data preprocessing, exploratory analysis, model training, evaluation, and result visualization.

Future improvements can include advanced algorithms such as Random Forest, XGBoost, and Gradient Boosting to improve prediction accuracy.